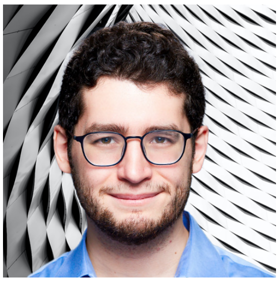
Tyler Barkin · Consultant and Embedded Engineer · Oyster Bay, New York · **tybarkin@gmail.com** · (516) 232-6378

Tyler Barkin | Resume



- Status: M.Sc. Engineering Robotics, B.Sc Mechanical Engineering
- Fields: Embedded Systems, Reinforcement Learning, Software Consulting
- Software: C, C++, C#, Python, PyTorch, TensorFlow, Solidworks, Altium, GIT, Jira/Confluence, ROS, Unity, UE5, Adobe Suite
- Hardware: CNC Lathe, CNC Mill, 3D Printers, Logic Analyzer, Function Generator, Hot Air Reflow, Oscilloscope, Solder Rework Station

Summary

I am a Robotics Engineer specializing in embedded software, simulation, and reinforcement learning. My expertise includes both hardware and software development, allowing me to prototype mechatronic systems from the ground up. I am looking for a role where my skills in embedded system development and edge-based AI can flourish. With a proven track record in embedded development and strong skills in problem-solving, physical simulation, and algorithm design, I am ready to be a valuable contributor to any engineering team.

Publications and Patents

2023 — **A Framework for Developing Robust, Autonomous, Power Managed Dynamic Soaring Flight Controllers Using Deep Reinforcement Learning** — AIAA Aviation and Aeronautics Forum

- I wrote the flight simulation algorithm in pytorch to take advantage of GPU parallelization. This strategy allows for simultaneous stimulation of thousands of environments, bringing policy training time from days to hours. The resulting control policy was a novel breakthrough, representing one of the first dynamic soaring flight controllers capable of flying in any arbitrary direction using only the power of the wind.
- Additional flight strategies were developed under this framework. Powered and unpowered flight, fixed and variable pitch propellers, and even battery charging routines were all learned and successfully executed in simulation

2023 — **A Mobile Reinforcement Learning-Cyber-Physical Fluid Dynamics-based Flapping Wing Platform: Simulation Component** — AIAA SciTech 2023 Forum

- Used Isaac Gym to develop locomotion policies for an aquatic robot with articulated paddle actuators. This method produced some unique locomotion modes that were dependent on the target swim velocity.

2020 — **Needle-free injector with gas bubble detection** — Portal Instruments

- Engineered an algorithm to estimate cartridge pressure in a needle-free drug injection device, crucial for precise control of stream velocity. This innovation significantly enhanced the safety and effectiveness of the auto-injector by accurately managing fluid drill depth, thereby preventing patient harm.

Education

2012/ 14 — **Graduated as M.Sc. in Engineering Robotics** — University of Pennsylvania

2009/ 12 — **Graduated as B.Sc. in Mechanical Engineering** — Washington University in Saint Louis

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Experience

2022 - Pres **Independent Research** Vento AI

- Designing and building a 3D printed RC aircraft that is capable of executing neural network control policies. Simulation software was written with pytorch for GPU parallelization of algorithms
- Working on a quadruped robot. Developing all the mechanical, electrical and software components required to make a robot walk naturally. Leveraging Isaac Gym as a simulation environment.

2021 - Pres **Embedded Engineer** RevMedica

- Wrote drivers to integrate accelerometers, motor drivers, light sensors, encoders and a TFT display for RevMedica's laparoscopic stapling device.
- Optimized existing software by developing non-blocking communication protocols. This change allowed for a more stable control loop timing

2022 - 2022 **Firmware Engineer** ULC Robotics

- Improved an audio signal detection algorithm by replacing the traditional Discrete Fourier Transform (DFT) and correlation algorithm with a Recursive Neural Network (RNN). This upgrade resulted in a remarkable improvement in detection sensitivity and a reduced loop execution time from 12 seconds to 0.5 seconds, significantly enhancing the system's accuracy and efficiency.
- Responsible for maintaining and upgrading firmware for ULCs embedded systems. Successfully transitioned the development environment from STM32Cube IDE to a CLion Makefile-based system, substantially optimizing the development workflow.

2021 - 2021 **Teaching Assistant: Introduction to Deep Learning** Harvard Extension School

- Designed comprehensive lesson plans and led engaging discussions to clarify complex concepts.
- Developed and provided practical coding examples to reinforce theoretical knowledge, contributing significantly to student understanding and success.

2019 - 2021 **GNC Engineer II** AeroVironment

- Led the software engineering development for a camera stabilization gimbal on a small UAV. Conducted detailed vibration analysis to identify and mitigate resonant frequencies, aiding in damper selection and enhanced stability.
- Designed and fabricated a PCB development board that interfaced with a Xilinx FPGA. This board broke out all essential sensors and served as a foundational platform for software development

2015 - 2019 **Systems Engineer** Portal Instruments

- Pioneered a method to accurately determine gas bubble volume in drug cartridges during injections, critical for maintaining precise injection velocity profiles. This innovation ensured effective drug delivery to subcutaneous tissues and led to a patent.